Neural_Networks

LSTM

1. $\mathbf{i}_t = \sigma(\mathbf{W}_{ix}\mathbf{x}_t + \mathbf{W}_{ih}\mathbf{h}_{t-1})$,
2. $\mathbf{f}_t = \sigma(\mathbf{W}_{fx}\mathbf{x}_t + \mathbf{W}_{fh}\mathbf{h}_{t-1})$,
3. $\mathbf{o}_t = \sigma(\mathbf{W}_{ox}\mathbf{x}_t + \mathbf{W}_{oh}\mathbf{h}_{t-1})$,
4. $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_{cx}\mathbf{x}_t + \mathbf{W}_{ch}\mathbf{h}_{t-1})$,
5. $\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$,

References:

1. LSTM: A detailed Explanation, Towards Data Science, Rian Dolphin.

Self-Supervised Learning

References:

1. Unsupervised Representation Learning by Predicting Image Rotations. Gidaris, et. al. Ecole des Ponts ParisTech.